Lecture 3 — Before class

Introduction to Quantitative Biology | Read before Thursday

In Lecture 2 we met a gene that **represses** itself, and saw that it reaches its steady state faster and holds it more steadily than a gene without feedback. Lecture 3 asks what happens when a gene **activates** itself. The answer is a new behaviour: a cell that can sit in one of two states, and remember which.

This sheet is short on purpose. Before Thursday:

1. **Read Section 1** below (one page): how to read a differential equation from a picture, without solving it.
2. **Read Section 5.1.1 of Alon's book**, pp. 78–80 (three pages).
3. **Write down your answers to the questions in Section 3**, and bring them.

1. Reading an equation without solving it

Every model of one gene we have written so far has the same form,

$$\frac{dX}{dt} = f(X) - \alpha X,$$

production minus removal. In Lecture 1 f was a constant and we solved the equation. When f depends on X we usually cannot — but most of what we want to know can be read from a picture.

Draw the two terms against X on the same axes: the production $f(X)$ and the removal αX, which is a straight line through the origin (Figure 1).

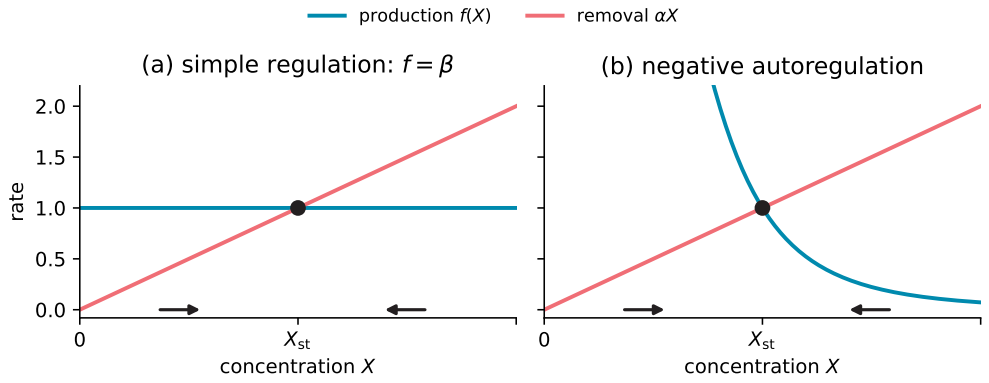


Figure 1: The rate plot. (a) A gene without feedback: the crossing is the steady state $X_{\text{st}} = \beta/\alpha$ of Lecture 1, found here without solving anything. (b) A gene that represses itself, as in Lecture 2. In both cases the curves cross once and the arrows point towards the crossing: one stable fixed point.

The picture is read with three rules:

1. **Where production is above removal**, $dX/dt > 0$ and X increases: draw an arrow pointing right on the X axis. Where production is below removal, X decreases: arrow pointing left.

2. **Where the two curves cross**, $dX/dt = 0$ and X stops changing. Such a point is called a **fixed point**; it is a steady state.
3. A fixed point is **stable** if the arrows on both sides point towards it: push X a little away, and it comes back. It is **unstable** if the arrows point away from it: the smallest push, and X runs off.

The gap between the curves is a speed

The vertical distance between production and removal is $|dX/dt|$, the speed at which X moves. Compare the two panels away from X_{st}: the gap is larger in (b). This is the picture behind the result of Lecture 2 that negative autoregulation responds faster.

2. Reading

Required: Alon, **Section 5.1.1**, "Positive autoregulation slows responses and can lead to bistability", pp. 78–80 (Alon, *An Introduction to Systems Biology*, 2nd ed., 2019). It uses exactly the picture of Section 1.

Optional, for those who want more

- Alon, **Section 5.1.2** (two genes regulating each other, the toggle switch) — where the course goes next.
- Strogatz, *Nonlinear Dynamics and Chaos* (2nd ed., 2015), **Sections 2.0–2.2** (pp. 15–20): the same rules as Section 1, in general, with "flows on the line". **Section 2.4** (pp. 24–26): how fast a stable point attracts, from its slope.
- Strogatz's own lectures, filmed at Cornell in 2014 and following the book, on YouTube (lectures 1 and 2 cover the above): https://www.youtube.com/playlist?list=PLbN57C5Zdl6j_qJA-pARJnKsmROzPnO9V.

3. Questions to bring on Thursday

Exercise 1: Write down your answers

In these questions f is the activating input function of the Lecture 2 pre-read, plus a small **basal production** β_0: the rate at which the gene is made even when no X is bound.

1. Self-activation, $n = 1$**.** Take $f(X) = \beta_0 + \beta\, X/(K + X)$. Sketch the rate plot. How many times can $f(X)$ cross the line αX? Is the crossing stable?

2. Self-activation, $n = 4$**.** Now $f(X) = \beta_0 + \beta\, X^4/(K^4 + X^4)$. Sketch it. Can it cross the line three times? If so, mark which fixed points are stable and which is not.

3. Memory. In the situation of question 2, a cell sits at the lowest fixed point. A signal briefly increases the production of X and pushes it past the middle fixed point; then the signal disappears. Where does the cell end up? Alon gives one example where this matters — which one?

4. Faster or slower? Does a gene that activates itself reach its steady state faster or more slowly than a gene without feedback? Answer using the gap between the curves.